

Alya Karim

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Builds REST APIs, AI-powered systems, and data pipelines in Python, Java, and React, with a 1:1 MSc in Data Analytics. Translates complex requirements into clean, production-ready solutions, integrating AI into everyday development workflows to ship faster and maintain higher code quality. Based in Ireland on a Stamp 1G visa.

TECHNICAL SKILLS

Languages & Core: Python, TypeScript, JavaScript, Java, SQL, R, HTML, CSS

Backend & Data: FastAPI, SQLAlchemy, PostgreSQL, REST APIs, JWT, Docker, Git, PyTest, Spark, Snowflake, Databricks, dbt, pandas, PyTorch, OpenCV, LangChain, FAISS, OpenAI API

Frontend & Tools: React, Vite, Tailwind CSS, Stripe API, Supabase, Vercel, Power BI, Tableau, Postman, Jira, Agile Scrum

PROJECTS

TaskFlow REST API | Python, FastAPI, SQLAlchemy, PostgreSQL, PyTest | github.com/alymkarim/taskflow 2026

- Built and shipped a task management REST API using FastAPI and SQLAlchemy, implementing full CRUD, JWT authentication, role-based access control, input validation, and API documentation, with a modular architecture designed for clean production migration.
- Actively developing new features and writing unit tests with PyTest throughout, applying AI-assisted development workflows to iterate quickly, catch issues earlier, and maintain clean, modular code throughout.

CartOS: Payment Platform | React, TypeScript, FastAPI, Stripe, PostgreSQL, Supabase, JWT | In Development 2026

- Building a full-stack payment platform with a FastAPI backend handling user authentication, product management, checkout, and orders, integrating Stripe for payment processing and real-time payment confirmation, storing all order data in PostgreSQL, and a React and TypeScript frontend with product catalogue, checkout flow, and order history.
- Built with security and production readiness in mind throughout: protected endpoints, secure password hashing, input validation, and structured API documentation, targeting Docker containerisation and cloud deployment.

DeepGuard: Real-Time Deepfake Detection | Python, PyTorch, EfficientNet, OpenCV, MediaPipe, FastAPI | In Development 2026

- Building a computer vision inference pipeline using EfficientNet transfer learning and MediaPipe face detection, exposing real-time analysis via a FastAPI backend with Docker containerisation, targeting reliable deployment with low-latency inference.

ResearchIQ: Multi-Document RAG System | Python, FastAPI, LangChain, FAISS, OpenAI API 2025

- Built an AI knowledge system that lets users upload multiple PDFs and ask questions across them, using OpenAI embeddings and vector search to retrieve relevant content and return accurate, source-based answers via a FastAPI backend.

UrbanTech Co-Working Spaces | Ericsson Industry Project, Java, Maven, Tomcat, Jira 2025

- Collaborated within an Agile Scrum team of five for an Ericsson industry project, owning the login and booking module from requirements through unit testing and go-live, managing sprint tasks and defect tracking in Jira, and communicating progress clearly to team members and stakeholders throughout delivery.

WORK EXPERIENCE

Freelance Data Analyst | Self-Employed 2026 – present

- Delivered data analytics and reporting work independently, writing Python and SQL pipelines, producing structured outputs for non-technical clients, and using AI-assisted development workflows to work faster and ship cleaner code across multiple concurrent engagements.

Research Data Analyst | Technological University of the Shannon, Ireland 2021 – 2023

- Built and maintained analytical pipelines in Python and SQL, produced structured reports for senior stakeholders, and collaborated cross-functionally to keep data quality and delivery standards consistent throughout the programme.

EDUCATION

MSc in Data Analytics – 1:1 2024 – 2025

Technological University of the Shannon, Ireland

BSc (Hons) Applied Physics – 2:1 2016 – 2020

Universiti Teknologi PETRONAS, Malaysia

CERTIFICATIONS & PROFESSIONAL DEVELOPMENT

AWS Cloud Practitioner Essentials – AWS Skill Builder 2026

Associate Enterprise Data Analyst – CADS AI 2022